ZAYAN AHMAD GHOUS

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PROFESSIONAL SUMMARY

BSCS student (4th semester) with hands-on experience in Machine Learning, Data Engineering, and MLOps, specializing in model deployment, data preprocessing pipelines, and REST API development. Skilled in Python, TensorFlow, FastAPI, and cloud-based ML applications. Seeking a paid internship opportunity to contribute technical expertise while gaining industry experience.

TECHNICAL SKILLS

- **Programming Languages:** Python, C++, C
- **Machine Learning & AI:** TensorFlow, Keras, scikit-learn, NumPy, Pandas, Convolutional Neural Networks (CNN), classification, regression, model optimization, cross-validation
- **Data Engineering:** Data preprocessing, feature engineering, exploratory data analysis (EDA), data cleaning, feature scaling, categorical encoding, data augmentation
- **API Development:** FastAPI, Django REST Framework, REST API design, HTTP methods, API testing (Postman), Gradio
- **Tools & Platforms:** Git, GitHub, Jupyter Notebook, Google Colab, Hugging Face Spaces, Streamlit, Linux terminal
- **Data Processing Tools:** Pandas (data manipulation), NumPy (numerical operations), Matplotlib (data visualization)
- **Currently Learning:** Docker, MLOps pipelines, Kubernetes basics, LLM integration, advanced API frameworks

PROFESSIONAL EXPERIENCE

Machine Learning Engineer Intern University of Engineering & Technology (UET) Lahore **February 2025 – April 2025**

- Developed and trained multiple machine learning models including Decision Trees, Random Forests, and Neural Networks using Python and scikit-learn, achieving 85%+ accuracy on real-world datasets
- Built end-to-end data preprocessing pipelines including missing value handling, feature scaling, categorical encoding, and data transformation for datasets with 10,000+ samples
- Worked on data engineering tasks including data cleaning, ETL workflows, dataset integration, and automated preprocessing pipelines using Pandas and NumPy
- Designed CNN-based image classification models for MRI brain scan analysis using TensorFlow/Keras with data augmentation techniques to improve model robustness
- Deployed machine learning applications using FastAPI and Gradio for real-time inference and interactive web-based predictions
- Performed exploratory data analysis (EDA), feature engineering, and model evaluation while managing projects with Git/GitHub and Jupyter Notebook

PROJECTS

Brain Tumor Detection (CNN + FastAPI + Deployment) **Live:** huggingface.co/spaces/Ghous/brain_tumor

- Built a CNN-based brain tumor classification model achieving 88% accuracy on MRI dataset (500+ images) using TensorFlow/Keras
- Improved model generalization using data augmentation, dropout regularization, batch normalization, and early stopping
- Developed REST API using FastAPI for real-time inference and integrated with a Gradio UI for user-friendly predictions
- Deployed end-to-end ML application on Hugging Face Spaces for public web access
- Tech Stack: Python, TensorFlow, Keras, FastAPI, Gradio, scikit-learn

Diabetes Prediction System (End-to-End ML Pipeline) **Live:** huggingface.co/spaces/Ghous/diabetes-prediction

- Built a diabetes risk prediction model using scikit-learn achieving 92% accuracy on UCI dataset
- Performed full ML pipeline: data preprocessing, feature scaling (StandardScaler), EDA, and feature analysis using Pandas and NumPy
- Developed interactive Gradio + FastAPI application for real-time predictions via web interface
- Deployed production-ready ML app on Hugging Face Spaces with API integration and error handling
- Tech Stack: Python, scikit-learn, Pandas, NumPy, FastAPI, Gradio

EDUCATION

Board of Intermediate and Secondary Education **Lahore, Pakistan**
Intermediate in Computer Sciences (Punjab Group of colleges) 2022 – 2024
Minhaj University Lahore **Lahore, Pakistan**
BS in Computer Sciences 2024 – Present